On-Device Multimodal Health Reasoning Under Resource Constraints Using CRMM-R

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Abstract- Medical Visual Question Answering (Medical VQA) has emerged as an important multimodal learning task that combines medical image understanding with natural language reasoning to support clinical decision-making. While recent vision-language models have demonstrated remarkable performance, most existing approaches rely on computationally intensive architectures that limit their applicability in resource-constrained environments such as mobile healthcare devices, edge computing platforms, and low-resource clinical settings. Furthermore, relatively few studies jointly investigate predictive performance, calibration reliability, robustness against input perturbations, and computational efficiency within a unified evaluation framework.

This paper presents **CRMM-R (Cross-Modal Resource-efficient Multimodal Reasoning)**, a lightweight multimodal framework designed for on-device medical reasoning under resource constraints. The proposed architecture employs a convolutional neural network for visual feature extraction, a gated recurrent unit for textual representation learning, and a cross-attention fusion mechanism to effectively align image and question representations before answer prediction. In addition to improving multimodal interaction, the framework systematically evaluates robustness through input corruption experiments, confidence reliability using Expected Calibration Error (ECE), deployment efficiency via dynamic quantization, and computational characteristics including model size, inference latency, throughput, and parameter complexity. Experimental evaluation on the **VQA-RAD** dataset demonstrates that the proposed cross-attention architecture improves prediction accuracy from **28.20%** to **35.81%** compared with the baseline concatenation model while maintaining an Expected Calibration Error of **0.0437**, reducing the quantized model size to **0.90 MB**, and sustaining efficient inference suitable for resource-limited environments. Failure analysis further identifies limitations associated with visual grounding and modality bias, providing insights for future multimodal reasoning research. The results demonstrate that carefully designed lightweight cross-modal fusion architectures can achieve an effective balance between predictive performance, reliability, and computational efficiency, making them well suited for practical deployment in edge-enabled medical AI systems.

Keywords: *Medical Visual Question Answering (Medical VQA), Multimodal Learning, Cross-Attention Fusion, Vision-Language Models, Resource-Constrained Artificial Intelligence, Edge AI, Medical Image Analysis, Model Calibration, Expected Calibration*

Error (ECE), Noise Robustness, Dynamic Quantization, Model Compression, On-Device Intelligence, Deep Learning, Healthcare Artificial Intelligence.

I. INTRODUCTION

The rapid advancement of **Artificial Intelligence (AI)** has significantly transformed medical image analysis by enabling automated interpretation, diagnosis support, and clinical decision-making across diverse healthcare applications. Among these developments, **Medical Visual Question Answering (Medical VQA)** has emerged as an important multimodal reasoning task that requires simultaneous understanding of medical images and natural language questions to generate clinically meaningful answers. Unlike conventional image classification or object detection, Medical VQA integrates visual perception with semantic reasoning, allowing intelligent systems to assist healthcare professionals by answering diagnostic, anatomical, and pathological questions based on medical imaging data. Such systems have the potential to improve clinical workflow efficiency, facilitate medical education, and enhance decision support, particularly in environments with limited access to experienced specialists.

Despite considerable progress in **vision-language models**, existing Medical VQA systems are predominantly built upon computationally intensive architectures, including large transformer-based models and deep multimodal networks containing millions or even billions of parameters. Although these architectures achieve high predictive performance, their substantial computational requirements, memory consumption, and inference latency limit their deployment on **resource-constrained platforms**, including mobile healthcare devices, embedded diagnostic systems, edge computing environments, and rural healthcare infrastructures. Consequently, designing lightweight multimodal reasoning models capable of maintaining reliable predictive performance while operating under strict computational constraints has become an increasingly important research direction.

A fundamental challenge in lightweight Medical VQA lies in the effective integration of heterogeneous information obtained from medical images and textual queries. Traditional multimodal architectures frequently employ simple feature concatenation or independent modality processing, resulting in insufficient interaction between visual and textual representations. Such limited cross-modal communication often produces weak semantic alignment, reduced reasoning capability, and an increased dependence on textual priors rather than clinically relevant visual evidence. Furthermore, the reliability of many existing systems is typically evaluated using prediction accuracy alone, overlooking equally important characteristics such as confidence calibration, robustness against noisy inputs, computational efficiency, and deployment feasibility under constrained hardware environments.

Recent advances in **cross-attention mechanisms** have demonstrated the ability to improve multimodal representation learning by enabling dynamic interaction between image and language features. Instead of treating visual and textual information independently, cross-attention allows each modality to selectively emphasize information from the other, thereby improving contextual understanding and semantic consistency. While these mechanisms have achieved promising results in large-scale vision-language models, comparatively limited research has investigated their effectiveness within lightweight architectures specifically designed for on-device medical reasoning.

Motivated by these observations, this research investigates the following question:

How can lightweight cross-modal fusion improve predictive performance, calibration reliability, robustness, and computational efficiency in Medical Visual Question Answering under resource-constrained environments?

To address this problem, this paper proposes **CRMM-R (Cross-Modal Resource-efficient Multimodal Reasoning)**, a lightweight multimodal framework designed for efficient medical reasoning on computationally constrained devices. The proposed architecture integrates a convolutional neural network for visual feature extraction, a gated recurrent unit for textual representation learning, and a cross-attention fusion module to establish effective interaction between image and language representations before answer prediction. Beyond architectural improvements, the framework presents a comprehensive experimental evaluation that examines predictive accuracy, robustness against input perturbations, confidence calibration using **Expected Calibration Error (ECE)**, model compression through dynamic quantization, and computational characteristics including model size, inference latency, throughput, and parameter complexity. This comprehensive evaluation provides a holistic assessment of model behavior beyond conventional accuracy-based comparisons.

The principal contributions of this research are summarized as follows:

- **A lightweight multimodal reasoning framework (CRMM-R)** is proposed for Medical Visual Question Answering under resource-constrained environments.
- **A cross-attention fusion mechanism** is introduced to improve semantic interaction between visual and textual representations while maintaining computational efficiency.
- **A comprehensive evaluation framework** is established by jointly analyzing predictive accuracy, robustness, calibration, compression, computational efficiency, and failure modes.
- **Extensive experiments on the VQA-RAD dataset** demonstrate that CRMM-R consistently improves multimodal reasoning performance over the baseline concatenation architecture while preserving efficient on-device deployment characteristics.
- **Failure analysis and resource constraint evaluation** provide additional insights into model limitations, deployment feasibility, and future research opportunities for lightweight medical vision-language systems.

The remainder of this paper is organized as follows. Section II reviews existing literature on Medical Visual Question Answering, multimodal fusion, calibration, robustness, and resource-constrained artificial intelligence. Section III formulates the research problem and presents the mathematical framework. Section IV describes the proposed CRMM-R architecture and methodology. Section V details the experimental setup, while Section VI presents quantitative and qualitative experimental results. Section VII discusses the key findings, followed by limitations and future research directions in Section VIII. Finally, Section IX concludes the paper.

II. RELATED WORK

Recent advances in **Medical Visual Question Answering (Medical VQA)** have significantly expanded the application of multimodal artificial intelligence in healthcare by enabling systems to jointly interpret medical images and natural language queries. Unlike conventional computer vision tasks that focus solely on image classification or segmentation, Medical VQA requires integrated reasoning across heterogeneous modalities to generate clinically meaningful responses. The increasing availability of multimodal medical datasets and the rapid development of deep learning architectures have accelerated research in this area. However, achieving high predictive performance while maintaining computational efficiency and deployment feasibility remains an active research challenge.

A. Medical Visual Question Answering

Medical Visual Question Answering extends traditional Visual Question Answering by incorporating domain-specific medical knowledge into multimodal reasoning. Early Medical VQA systems primarily employed convolutional neural networks (CNNs) for visual feature extraction and recurrent neural networks (RNNs) or Long Short-Term Memory (LSTM) networks for question encoding, followed by simple feature concatenation for answer prediction. Although these approaches demonstrated the feasibility of multimodal medical reasoning, they frequently exhibited limited semantic interaction between image and textual representations, resulting in reduced reasoning capability for clinically complex questions.

More recent developments have introduced transformer-based vision-language models capable of learning richer contextual relationships between modalities through self-attention mechanisms. These architectures substantially improve representation learning and reasoning performance but often require extensive computational resources, large-scale pretraining datasets, and high-memory hardware accelerators. Consequently, their deployment on portable diagnostic devices, embedded healthcare systems, and resource-constrained clinical environments remains challenging. This limitation highlights the need for lightweight multimodal architectures capable of delivering reliable performance under practical deployment constraints.

B. Multimodal Fusion Methods

Effective multimodal reasoning depends largely on the strategy used to combine visual and textual information. Conventional fusion approaches frequently employ **feature concatenation**, where independently extracted image and text features are merged into a single representation before classification. While computationally efficient, concatenation assumes equal contribution from both modalities and provides limited capability for modeling semantic relationships between visual evidence and language.

To overcome these limitations, advanced fusion strategies such as bilinear pooling, co-attention, and cross-attention mechanisms have been proposed. Bilinear fusion captures higher-order interactions between modalities but introduces additional computational complexity. Co-attention jointly learns attention distributions across image and text representations, enabling stronger contextual alignment. More recently, cross-attention has emerged as an effective mechanism for multimodal representation learning by allowing one modality to selectively attend to relevant information from the other. This dynamic interaction improves semantic correspondence between visual features and linguistic context while avoiding the representational limitations associated with independent feature processing. Nevertheless, relatively few studies have investigated the application of lightweight cross-attention architectures for Medical VQA under strict computational constraints.

C. Model Calibration in Medical Artificial Intelligence

Prediction accuracy alone is insufficient for evaluating clinical decision-support systems, where the reliability of confidence estimates is equally important. **Model calibration** measures the degree to which predicted confidence reflects actual prediction correctness. Poorly calibrated models may assign high confidence to incorrect predictions, increasing the risk of unreliable clinical recommendations.

Expected Calibration Error (ECE) has become one of the most widely adopted metrics for quantifying confidence reliability in deep learning models. Recent research demonstrates that modern neural networks often exhibit overconfidence despite achieving competitive classification accuracy. Consequently, calibration analysis has become an essential component of trustworthy medical artificial intelligence. Despite its growing importance, calibration remains relatively underexplored in lightweight Medical VQA architectures, motivating its inclusion as a core evaluation criterion in this work.

D. Robustness Under Input Corruptions

Medical imaging systems frequently encounter input variations arising from acquisition noise, motion artifacts, resolution degradation, and equipment-specific inconsistencies. Robust multimodal reasoning systems must therefore maintain stable predictive performance under such perturbations. Existing robustness studies commonly evaluate deep learning models using synthetic corruptions including Gaussian noise, blur, contrast variation, and compression artifacts to simulate realistic imaging conditions.

Although transformer-based multimodal models generally demonstrate improved robustness through large-scale pretraining, lightweight architectures remain comparatively vulnerable to degraded image quality because of their reduced representational capacity. Understanding how multimodal reasoning models respond to input corruption is therefore essential for assessing deployment readiness in practical healthcare environments.

E. Resource-Constrained Multimodal Learning

The growing demand for intelligent healthcare applications on mobile devices, embedded systems, and edge computing platforms has stimulated extensive research into **resource-constrained artificial intelligence**. Lightweight neural architectures, model pruning, knowledge distillation, quantization, and efficient attention mechanisms have become widely adopted techniques for reducing computational complexity while preserving predictive performance.

Dynamic quantization has emerged as an effective post-training optimization strategy that substantially reduces model size and inference latency without requiring additional retraining. Similarly, lightweight convolutional networks combined with compact sequence encoders provide favorable trade-offs between computational efficiency and representation quality. However, existing studies generally emphasize either efficiency or predictive accuracy independently, with relatively limited attention given to jointly evaluating computational efficiency, robustness, calibration, compression sensitivity, and multimodal reasoning performance within a unified experimental framework.

Summary of Research Gap

The literature indicates substantial progress in Medical Visual Question Answering, multimodal fusion, model calibration, and efficient neural network design. Nevertheless, three major research gaps remain evident:

1. Existing lightweight Medical VQA systems often rely on simple feature fusion strategies that provide limited cross-modal semantic interaction.
2. Most studies prioritize predictive accuracy while overlooking complementary evaluation dimensions such as calibration, robustness, computational efficiency, and deployment feasibility.
3. Comprehensive investigations that simultaneously analyze multimodal reasoning quality and resource-constrained deployment characteristics remain relatively scarce.

To address these limitations, the proposed **CRMM-R** framework integrates lightweight cross-attention fusion with systematic evaluation of predictive performance, calibration reliability, robustness under input perturbations, model compression, computational efficiency, and failure behavior, thereby providing a comprehensive framework for on-device multimodal health reasoning under resource constraints.

III. Problem Formulation

Medical Visual Question Answering (Medical VQA) is a multimodal learning task that requires a model to interpret a medical image together with a natural language question and generate the most appropriate clinical answer. Unlike traditional image classification tasks, Medical VQA demands simultaneous understanding of visual content and linguistic context, making it a challenging reasoning problem. The objective is to accurately associate medical image features with the semantic meaning of the corresponding question while producing reliable and clinically relevant predictions.

The proposed CRMM-R framework addresses this problem by learning meaningful representations from both medical images and textual questions before integrating them through a cross-modal reasoning mechanism. The system is designed to improve semantic alignment between the two modalities, enabling more accurate answer prediction while maintaining computational efficiency suitable for deployment on resource-constrained devices.

A. Problem Definition

The primary objective of this research is to develop a lightweight multimodal reasoning framework capable of answering clinical questions using information extracted from medical images. Each input sample consists of a medical image, an associated natural language question, and the corresponding ground-truth answer. During inference, the framework receives the image-question pair as input and predicts the most appropriate answer from a predefined answer vocabulary.

Unlike conventional Medical VQA systems that emphasize predictive accuracy alone, this work considers multiple performance dimensions to evaluate the practical effectiveness of the proposed architecture.

B. Research Objectives

The objectives of this study are summarized as follows:

- Design a lightweight multimodal architecture for Medical Visual Question Answering.
- Improve semantic interaction between image and textual representations using a cross-attention fusion mechanism.
- Evaluate the robustness of the model under noisy imaging conditions.
- Assess confidence reliability through calibration analysis using Expected Calibration Error (ECE).
- Investigate the impact of model compression on predictive performance.
- Analyze computational efficiency using parameter count, model size, inference latency, and throughput.

- Examine model failure patterns to identify limitations and opportunities for future improvements.

C. Resource Constraints

The proposed CRMM-R framework is specifically designed for deployment in environments where computational resources are limited. The system emphasizes efficient memory utilization, low computational complexity, and reduced storage requirements while maintaining acceptable predictive performance. Resource efficiency is evaluated using multiple deployment-oriented metrics, including the number of trainable parameters, model size, quantized model size, inference time, and throughput. These metrics collectively determine the feasibility of deploying the framework on edge devices, embedded medical systems, and portable healthcare platforms.

D. Research Scope

This study focuses on investigating the effectiveness of lightweight cross-modal reasoning for Medical Visual Question Answering using the VQA-RAD dataset. The evaluation encompasses not only prediction accuracy but also robustness under image corruption, confidence calibration, compression sensitivity, computational efficiency, and failure analysis. The study does not attempt to replace large-scale foundation models; instead, it aims to demonstrate that carefully designed lightweight architectures can provide a practical balance between predictive performance and resource efficiency for real-world medical AI applications.

IV. Proposed Methodology

The proposed **Cross-Modal Resource-efficient Multimodal Reasoning (CRMM-R)** framework is designed to perform Medical Visual Question Answering (Medical VQA) while satisfying the computational constraints associated with on-device deployment. The framework integrates visual feature extraction, textual representation learning, cross-modal semantic fusion, and lightweight classification into a unified architecture capable of reasoning over heterogeneous medical information. Unlike conventional Medical VQA models that rely on independent feature processing or simple feature concatenation, the proposed architecture enables continuous interaction between visual and textual representations through a cross-attention mechanism, thereby improving multimodal reasoning while maintaining computational efficiency.

The complete CRMM-R pipeline consists of six sequential processing stages: **(1) image feature extraction, (2) question encoding, (3) multimodal feature fusion, (4) answer classification, (5) resource-aware optimization, and (6) prediction generation**.

A. Overall CRMM-R Framework

The CRMM-R framework follows an end-to-end multimodal learning pipeline that simultaneously processes medical images and natural language questions to generate clinically relevant answers. The architecture is specifically designed to maximize semantic interaction between visual and textual modalities while minimizing computational overhead, thereby enabling deployment in resource-constrained healthcare environments.

For each input sample, the system receives two complementary sources of information: a medical image obtained from the VQA-RAD dataset and a corresponding natural language question describing the diagnostic task. These two modalities contain different but interdependent information. The medical image provides anatomical and pathological visual evidence, whereas the textual question specifies the clinical context required for reasoning. Consequently, both modalities must be jointly analyzed before an appropriate answer can be predicted.

The first stage of the framework performs **visual representation learning**. The input medical image is resized to a uniform resolution of **128 × 128 pixels** before being forwarded to a lightweight Convolutional Neural Network (CNN). The image encoder consists of multiple convolutional layers followed by Rectified Linear Unit (ReLU) activation functions and max-pooling operations that progressively extract hierarchical visual features while reducing spatial redundancy. An Adaptive Average Pooling layer converts the final feature maps into a fixed-dimensional representation, allowing images of consistent size to be transformed into compact visual embeddings suitable for multimodal fusion.

Simultaneously, the natural language question undergoes **textual representation learning**. The input question is first preprocessed through tokenization, vocabulary mapping, and integer encoding. Each token is converted into a dense embedding vector before being processed by a Gated Recurrent Unit (GRU). The recurrent architecture captures sequential dependencies among words and produces a contextual representation that summarizes the semantic content of the entire question. Compared with larger transformer-based language encoders, the GRU provides a favorable balance between representational capability and computational efficiency, making it well suited for lightweight deployment.

After independent feature extraction, the visual and textual embeddings are forwarded to the **Cross-Modal Fusion Module**, which represents the principal contribution of the proposed architecture. Instead of simply concatenating the feature vectors, CRMM-R employs a cross-attention mechanism that enables each modality to selectively attend to relevant information contained within the other. During this process, textual representations guide the selection of discriminative visual features, while visual representations reinforce clinically meaningful contextual information contained within the question. This bidirectional interaction strengthens semantic alignment and improves multimodal reasoning without introducing the computational complexity associated with large transformer architectures.

The fused multimodal representation generated by the cross-attention module is subsequently passed through a lightweight classification network composed of fully connected layers with ReLU activation. The classifier transforms the fused feature representation into a probability distribution over the predefined answer vocabulary using a Softmax output layer. The answer corresponding to the highest predicted probability is selected as the final response generated by the Medical VQA system.

Beyond prediction accuracy, the proposed framework incorporates several design considerations to ensure practical deployment under computational constraints. Lightweight convolutional feature extraction, GRU-based language modeling, dynamic quantization, and efficient cross-attention collectively reduce memory consumption, model size, and inference latency while maintaining competitive reasoning performance. These characteristics enable CRMM-R to operate effectively on edge computing devices, embedded medical systems, and portable diagnostic platforms where computational resources are inherently limited.

B Dataset Description

The effectiveness of any Medical Visual Question Answering (Medical VQA) system is fundamentally dependent on the quality and diversity of the dataset used for training and evaluation. To assess the proposed CRMM-R framework, experiments were conducted using the **VQA-RAD (Visual Question Answering–Radiology)** dataset, a publicly available benchmark specifically designed for multimodal reasoning in medical imaging. Unlike conventional computer vision datasets that primarily emphasize object recognition or scene understanding, VQA-RAD requires simultaneous interpretation of radiological images and natural language questions, thereby providing an appropriate benchmark for evaluating cross-modal reasoning architectures.

The VQA-RAD dataset consists of **315 medical image–question–answer triplets**, where each sample contains a radiological image, an associated clinical question, and the corresponding ground-truth answer. Every sample represents a complete multimodal reasoning instance in which the model must integrate visual evidence with linguistic information to generate the correct response. This structured format closely resembles real-world clinical scenarios in which physicians interpret diagnostic images while answering questions related to anatomical structures, pathological findings, imaging modalities, and diagnostic observations.

A distinguishing characteristic of VQA-RAD is its inclusion of diverse **radiological imaging modalities**, enabling the evaluation of multimodal reasoning across different diagnostic contexts. The dataset includes **X-ray**, **Computed Tomography (CT)**, and **Magnetic Resonance Imaging (MRI)** scans, each presenting unique visual characteristics and clinical interpretation challenges. X-ray images emphasize skeletal structures and thoracic abnormalities, CT scans provide cross-sectional anatomical information with enhanced structural detail, while MRI images offer high soft-tissue contrast for neurological and musculoskeletal analysis. The inclusion of multiple imaging modalities increases the complexity of the learning task and enables comprehensive assessment of the proposed framework across heterogeneous medical imaging domains.

The accompanying clinical questions cover multiple categories of medical reasoning rather than simple object identification. These include anatomical localization, abnormality detection, imaging modality recognition, organ identification, pathological assessment, spatial reasoning, and binary diagnostic decision-making. Consequently, the model must simultaneously understand image content, interpret clinical terminology, and establish semantic correspondence between the two modalities before generating an appropriate answer. This characteristic makes VQA-RAD substantially more challenging than traditional image classification datasets.

Unlike open-ended natural language generation tasks, the VQA-RAD dataset defines a finite **answer vocabulary**consisting of unique textual responses collected from all annotated samples. During preprocessing, duplicate answers are removed, and each distinct answer is assigned a unique numerical class label through label encoding. The Medical VQA task is therefore formulated as a **multi-class classification problem**, where the model predicts the most probable answer from the predefined answer vocabulary. This formulation simplifies model optimization while maintaining clinically meaningful evaluation.

Prior to model training, the dataset undergoes structured preprocessing to standardize both visual and textual information. Medical images are resized to a fixed spatial resolution to ensure consistent feature extraction, while natural language questions are cleaned, tokenized, and converted into numerical sequences suitable for neural network processing. The resulting multimodal dataset provides aligned image-question-answer triplets that serve as direct input to the proposed CRMM-R architecture.

The selection of VQA-RAD is motivated by several practical considerations. First, it is one of the most widely adopted benchmark datasets for Medical Visual Question Answering, facilitating meaningful comparison with existing multimodal architectures. Second, its multimodal structure closely reflects real clinical reasoning tasks encountered in diagnostic practice. Finally, the relatively compact dataset size provides an appropriate benchmark for evaluating lightweight multimodal architectures designed for resource-constrained deployment, where efficient learning from limited annotated medical data is essential.

C Data Preprocessing

Data preprocessing plays a crucial role in multimodal learning by ensuring that heterogeneous data obtained from different modalities can be processed efficiently by the neural network. Since medical images and natural language questions differ significantly in representation and dimensionality, separate preprocessing pipelines are employed for the visual and textual modalities before feature extraction. The objective of this stage is to standardize the input data, reduce variability, and generate consistent numerical representations that facilitate effective multimodal learning.

1) Image Preprocessing

Medical images obtained from the VQA-RAD dataset exhibit variations in spatial resolution and image dimensions. To ensure uniformity across all training samples, each image is resized to a fixed resolution of **128 × 128 pixels** before being processed by the convolutional neural network.

The resizing operation offers several advantages. First, it guarantees that all input images possess identical spatial dimensions, enabling efficient batch processing during model training. Second, reducing the image resolution lowers computational complexity and memory consumption without significantly affecting the diagnostic information required for the lightweight CNN encoder. Finally, standardized image dimensions simplify feature extraction by maintaining a consistent input shape throughout the learning process.

After resizing, each image is converted into a tensor representation suitable for PyTorch-based processing. Pixel intensities are normalized to floating-point values within the interval **[0,1]**, ensuring numerical stability during optimization and facilitating faster convergence of the neural network.

2) Question Preprocessing

Clinical questions associated with medical images are expressed in natural language and therefore require transformation into numerical representations before being processed by the neural network. The preprocessing pipeline begins by converting all characters to lowercase, thereby eliminating inconsistencies caused by differences in capitalization.

Special characters, punctuation symbols, and unnecessary textual artifacts are subsequently removed using regular expression filtering, preserving only alphanumeric tokens relevant to semantic interpretation. The cleaned sentence is then segmented into individual words through whitespace-based tokenization.

Since neural networks require fixed-length numerical inputs, each token is mapped to a unique integer identifier using a predefined vocabulary. Questions containing fewer than the maximum sequence length are padded using a dedicated padding token, while longer questions are truncated to maintain a consistent input dimension. This procedure ensures uniform sequence lengths across the entire dataset and enables efficient mini-batch training.

3) Vocabulary Construction

A domain-specific vocabulary is constructed from all questions contained within the VQA-RAD dataset. Word frequencies are computed across the complete corpus, and the most frequently occurring terms are retained to form the final vocabulary.

Two reserved tokens are incorporated into the vocabulary to improve robustness during inference. The **<pad>** token is used for sequence padding, ensuring equal-length input sequences, while the **<unk>** token represents previously unseen or infrequently occurring words that are not included in the vocabulary. This strategy enables the model to generalize more effectively when encountering unfamiliar clinical terminology during prediction.

Each vocabulary entry is assigned a unique numerical index, allowing textual questions to be represented as sequences of integer identifiers suitable for embedding-based representation learning.

4) Answer Encoding

Unlike conventional language generation tasks, the proposed Medical Visual Question Answering framework formulates answer prediction as a **multi-class classification problem**. Every unique answer appearing within the dataset is treated as an independent output class.

During preprocessing, duplicate answers are removed to generate a unique answer vocabulary. Each answer is subsequently assigned a numerical class identifier through label encoding. Consequently, the neural network predicts an integer class label that is later mapped back to its corresponding textual answer during inference.

This formulation significantly simplifies optimization by enabling the use of categorical probability distributions and Cross-Entropy Loss while maintaining direct correspondence between predicted labels and clinically meaningful responses.

5) Data Loading Pipeline

Following preprocessing, the standardized images, encoded questions, and label-encoded answers are combined to construct the final multimodal dataset. A custom dataset class is implemented to efficiently retrieve synchronized image-question-answer triplets during training.

Mini-batch learning is performed using the PyTorch **DataLoader**, which automatically groups samples into batches of sixteen observations and randomly shuffles the training data before each epoch. Randomized batch generation improves statistical diversity during optimization and reduces the likelihood of learning order-specific patterns from the dataset.

The resulting preprocessing pipeline produces structured multimodal samples consisting of:

- A resized medical image represented as a tensor.
- A fixed-length encoded question sequence.
- A numerical answer label corresponding to the ground-truth response.

These standardized representations serve as the direct input to the CRMM-R architecture for multimodal feature extraction and reasoning.

D Image Encoder

The image encoder is responsible for extracting discriminative visual representations from medical images that can subsequently be integrated with textual information for multimodal reasoning. Since the objective of CRMM-R is to operate under resource-constrained environments, the image encoder is designed to provide an effective balance between representational capability and computational efficiency. Instead of employing computationally expensive deep convolutional backbones or transformer-based vision encoders, the proposed framework adopts a lightweight Convolutional Neural Network (CNN) architecture capable of learning hierarchical visual features while maintaining a compact model size and low inference latency.

Convolutional Neural Networks have demonstrated remarkable success in medical image analysis because of their ability to automatically learn spatial hierarchies directly from pixel intensities. Lower convolutional layers capture fundamental visual structures such as edges, gradients, and textures, whereas deeper layers progressively learn higher-level anatomical characteristics and pathological patterns. This hierarchical feature extraction process enables the network to preserve clinically relevant visual information while gradually reducing redundant spatial details.

The proposed image encoder consists of **three sequential convolutional blocks**, each designed to progressively transform the input image into a compact feature representation suitable for multimodal fusion.

The first convolutional block receives the resized **128 × 128 RGB medical image** and applies **32 convolutional filters**with a kernel size of **3 × 3**. Padding is employed to preserve spatial dimensions during convolution, allowing the network to maintain local anatomical information while extracting low-level visual features. Following convolution, the **Rectified Linear Unit (ReLU)** activation function introduces non-linearity into the network, enabling the model to learn complex feature representations that cannot be captured through linear transformations alone. A **Max Pooling** operation is subsequently applied to reduce the spatial resolution of the feature maps while retaining the most discriminative activations. This operation decreases computational complexity and improves translation invariance by preserving dominant anatomical structures.

The second convolutional block expands the feature representation by increasing the number of convolutional filters from **32 to 64**. As the receptive field increases, the encoder begins capturing more complex visual structures, including organ boundaries, tissue characteristics, and pathological regions. Similar to the previous stage, convolution is followed by ReLU activation and Max Pooling, allowing the network to progressively learn richer semantic information while reducing computational cost.

The third convolutional block further increases the feature dimensionality to **128 feature channels**. At this stage, the encoder focuses on extracting high-level medical representations that summarize the diagnostic content of the input image. Instead of performing additional pooling operations that could excessively reduce spatial information, the proposed architecture employs **Adaptive Average Pooling**, producing a fixed **4 × 4** feature map regardless of minor variations in intermediate feature dimensions. This design provides consistent feature representations while preserving sufficient spatial information required for multimodal reasoning.

The pooled feature maps are subsequently flattened into a one-dimensional feature vector and projected through a fully connected layer to generate a **256-dimensional visual embedding**. This embedding serves as a compact numerical representation of the medical image and forms the visual input to the Cross-Modal Fusion Module.

Several architectural decisions contribute to the computational efficiency of the proposed image encoder. First, the network contains only three convolutional stages, substantially reducing the number of trainable parameters compared with deeper architectures such as ResNet, DenseNet, or Vision Transformers. Second, Adaptive Average Pooling eliminates the need for multiple large fully connected layers, thereby reducing memory consumption. Finally, the resulting 256-dimensional feature representation provides sufficient semantic richness for multimodal reasoning while maintaining low computational overhead, making the architecture well suited for on-device deployment.

The choice of a lightweight CNN is motivated by both computational and practical considerations. Large transformer-based vision models generally require extensive computational resources, large-scale pretraining datasets, and high-performance GPUs to achieve competitive performance. Such requirements are often incompatible with mobile healthcare systems, embedded diagnostic devices, and edge computing platforms. In contrast, lightweight convolutional architectures offer lower inference latency, reduced memory usage, and faster execution while retaining strong feature extraction capability for relatively small medical datasets such as VQA-RAD. Consequently, the proposed CNN encoder provides an appropriate balance between predictive performance and deployment efficiency.

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E Cross-Modal Fusion Module

The integration of heterogeneous visual and textual information represents one of the most challenging aspects of Medical Visual Question Answering. Although independent image and language encoders can effectively learn modality-specific representations, accurate clinical reasoning requires meaningful interaction between these modalities. Conventional multimodal architectures frequently employ feature concatenation, where independently extracted image and text features are merged into a single vector before classification. While computationally simple, concatenation treats both modalities as independent sources of information and fails to explicitly model the semantic relationships that exist between clinical questions and corresponding medical images. Consequently, the classifier must learn cross-modal relationships implicitly, often resulting in weak semantic alignment and suboptimal reasoning performance.

To overcome this limitation, the proposed CRMM-R framework introduces a lightweight **Cross-Modal Attention Fusion Module** that enables direct interaction between visual and textual representations before answer prediction. Instead of merely combining feature vectors, the proposed module allows textual information to guide the selection of relevant visual evidence, thereby strengthening semantic correspondence between both modalities. This interaction improves multimodal reasoning while preserving the computational efficiency required for deployment in resource-constrained environments.

The fusion process begins with the visual feature vector generated by the CNN image encoder and the contextual textual representation produced by the GRU question encoder. Both representations are projected into a common feature space through independent linear transformation layers. These transformed representations are subsequently interpreted as **Query (Q)**, **Key (K)**, and **Value (V)** vectors that form the basis of the cross-attention mechanism.

Within the proposed architecture, the **textual feature vector serves as the Query**, while the **visual feature vector simultaneously serves as both the Key and the Value**. This design enables the semantic content of the clinical question to guide the selection of relevant visual information. In other words, the question determines which image features should receive greater emphasis during reasoning. Rather than processing every visual feature equally, the model dynamically allocates higher attention to image regions that are more relevant to the clinical query.

The attention mechanism computes similarity scores between the Query and Key representations to estimate the relevance of each visual feature with respect to the given question. These similarity scores are normalized using the Softmax function, producing attention weights that indicate the relative importance of different visual representations. The resulting attention distribution is subsequently applied to the Value representation, generating an attention-weighted visual feature vector that emphasizes diagnostically relevant image information while suppressing less informative features.

Unlike conventional feature concatenation, the proposed attention mechanism performs **adaptive feature selection**. Because the attention weights are computed separately for each image-question pair, the model dynamically adjusts its reasoning strategy according to the semantic requirements of the input question. Consequently, identical medical images may produce different attention distributions depending on the accompanying clinical query, enabling more flexible and context-aware multimodal reasoning.

To further improve representation learning, the proposed fusion module incorporates a **residual connection** between the original textual representation and the attention-weighted visual features. Rather than replacing the contextual information learned by the GRU, the residual connection combines both representations through element-wise addition. This design preserves the semantic information encoded within the clinical question while simultaneously enriching it with visually grounded evidence extracted from the medical image.

The fused representation produced by the Cross-Modal Attention Module therefore contains complementary information from both modalities. The textual representation provides contextual understanding of the clinical query, whereas the attention-weighted visual representation contributes anatomically relevant evidence. The resulting multimodal embedding forms a comprehensive representation that is subsequently forwarded to the classification network for answer prediction.

One of the principal advantages of the proposed fusion strategy is its computational efficiency. Unlike transformer-based cross-attention architectures that require multiple attention heads, deep stacking, and extensive matrix operations, the proposed module employs a lightweight attention mechanism consisting primarily of linear projections and element-wise operations. This substantially reduces parameter complexity and inference latency while preserving effective multimodal interaction. Consequently, the proposed architecture remains suitable for deployment on edge devices without sacrificing reasoning capability.

Compared with the baseline CRMM v1 architecture based on direct feature concatenation, the proposed cross-attention fusion mechanism demonstrates improved semantic alignment between visual and textual representations. Experimental evaluation presented in Section VI shows that replacing feature concatenation with cross-attention improves prediction accuracy from **28.20%** to **35.81%**, representing an absolute improvement of **7.61 percentage points** while maintaining efficient resource utilization. These results demonstrate that effective multimodal interaction can be achieved without relying on computationally intensive transformer architectures.

F Classification Layer

The final stage of the proposed CRMM-R architecture is the **classification layer**, which transforms the multimodal representation generated by the Cross-Modal Fusion Module into a clinically meaningful answer prediction. Following feature extraction and multimodal integration, the fused representation contains complementary visual and textual information that captures both the anatomical characteristics of the medical image and the semantic context of the associated clinical question. The objective of the classification layer is to utilize this integrated representation to determine the most probable answer from the predefined answer vocabulary.

The proposed classification network adopts a lightweight **fully connected feedforward architecture**, consistent with the overall design philosophy of CRMM-R. Rather than employing multiple deep classification blocks that substantially increase computational complexity, the proposed framework utilizes a compact two-layer neural classifier capable of performing efficient decision making while preserving deployment efficiency. This lightweight design reduces memory consumption and inference latency, making the framework suitable for execution on edge computing platforms and portable medical devices.

The fused multimodal feature vector is first forwarded to a **fully connected projection layer**, which reduces the feature dimensionality from **256** to **128** neurons. This dimensionality reduction encourages the network to learn compact discriminative representations while suppressing redundant information inherited from the preceding feature extraction modules. The reduced feature space also decreases the number of trainable parameters within the classifier, thereby improving computational efficiency without significantly compromising representational capacity.

A **Rectified Linear Unit (ReLU)** activation function is subsequently applied to introduce non-linearity into the classification process. The ReLU operation enables the classifier to learn complex decision boundaries that cannot be represented through linear transformations alone. Additionally, ReLU offers computational simplicity and efficient gradient propagation, making it particularly suitable for lightweight neural architectures intended for real-time inference.

The activated feature representation is then passed to the final **output layer**, which contains a number of neurons equal to the total number of unique answers present in the VQA-RAD dataset. Each output neuron represents the likelihood of a specific answer class. The network therefore formulates Medical Visual Question Answering as a **multi-class classification problem**, where each answer corresponds to an independent prediction category.

To convert the raw output scores into interpretable probabilities, the classifier employs the **Softmax activation function**. Softmax normalizes the outputs into a probability distribution whose values collectively sum to one. Consequently, each output value represents the predicted probability that the corresponding answer is the correct response to the given image-question pair. During inference, the answer associated with the highest predicted probability is selected as the final prediction produced by the CRMM-R framework.

Model optimization is performed using the **Cross-Entropy Loss** function, which measures the discrepancy between the predicted probability distribution and the ground-truth answer label. Cross-Entropy Loss is widely adopted for multi-class classification because it directly encourages higher probabilities for correct predictions while penalizing incorrect classifications. This optimization strategy enables efficient convergence during training and improves the discriminative capability of the classifier.

The lightweight design of the classification layer complements the overall resource-efficient architecture of CRMM-R. By combining compact fully connected layers with efficient activation functions and probabilistic output modeling, the classifier achieves accurate answer prediction while introducing only a minimal increase in computational complexity. Consequently, the classification module contributes to maintaining the overall efficiency of the framework without sacrificing predictive performance.

G Resource-Constrained Design

Resource efficiency has become a fundamental requirement for modern medical artificial intelligence systems, particularly in environments where computational resources, memory capacity, and storage availability are limited. Although recent multimodal transformer architectures have demonstrated impressive predictive performance, their deployment often requires high-end GPUs, extensive memory resources, and substantial computational power. Such requirements restrict their applicability in portable healthcare devices, embedded diagnostic systems, and edge computing platforms. The proposed CRMM-R framework is therefore designed with computational efficiency as a primary architectural objective, enabling practical deployment without substantially compromising multimodal reasoning capability.

The lightweight nature of CRMM-R is achieved through careful architectural design at every stage of the processing pipeline. The image encoder employs a compact three-layer convolutional neural network rather than computationally intensive deep residual or transformer-based architectures.

Similarly, the question encoder utilizes a Gated Recurrent Unit (GRU), which offers effective sequential representation learning while requiring significantly fewer parameters than transformer-based language models. The proposed Cross-Modal Attention Module further maintains computational efficiency by relying on lightweight linear projections and element-wise operations instead of multi-head transformer attention mechanisms.

Experimental evaluation demonstrates that the complete CRMM-R architecture contains approximately **1.37 million trainable parameters**, representing a compact model suitable for deployment on resource-limited hardware. The trained model occupies approximately **5.23 MB** of storage, substantially reducing memory requirements compared with large-scale multimodal foundation models that frequently exceed hundreds of megabytes or even several gigabytes. This compact model size facilitates rapid loading, efficient storage, and deployment on mobile and embedded healthcare devices.

To further improve deployment efficiency, **dynamic post-training quantization** is applied to the trained network. Dynamic quantization converts selected floating-point parameters into lower-precision integer representations without requiring additional retraining. Following quantization, the model size is reduced from **5.23 MB** to approximately **0.90 MB**, corresponding to a reduction of more than **80%** in storage requirements while preserving nearly identical predictive performance. This substantial reduction demonstrates the suitability of CRMM-R for memory-constrained deployment scenarios.

Inference efficiency is evaluated using CPU execution time and throughput measurements. Experimental results indicate an average **CPU inference time of approximately 353.89 milliseconds per batch**, while the framework achieves a throughput of approximately **45.21 images per second**. These results demonstrate that CRMM-R provides an effective balance between computational efficiency and predictive capability, supporting real-time or near-real-time inference for practical healthcare applications.

The computational characteristics of the proposed framework indicate that efficient multimodal reasoning can be achieved without relying on computationally expensive deep transformer architectures. The combination of lightweight convolutional feature extraction, compact sequential language modeling, efficient cross-modal attention, and post-training quantization collectively enables CRMM-R to satisfy the resource limitations associated with embedded diagnostic platforms, portable medical devices, and edge computing systems.

Unlike conventional Medical VQA systems that primarily optimize predictive accuracy, the proposed framework explicitly considers computational efficiency as an integral evaluation criterion. By jointly analyzing parameter complexity, storage requirements, inference latency, throughput, calibration reliability, robustness, and predictive performance, CRMM-R provides a comprehensive resource-aware multimodal reasoning framework that better reflects practical deployment requirements in real-world healthcare environments.

V. Experimental Setup

The proposed CRMM-R framework was evaluated through a comprehensive set of controlled experiments designed to assess its effectiveness in multimodal medical reasoning under resource-constrained environments. The experimental protocol was developed to evaluate not only predictive performance but also robustness, calibration reliability, computational efficiency, and deployment suitability. All experiments were conducted under identical training conditions to ensure fair comparison between the baseline architecture (CRMM v1) and the proposed cross-modal attention framework (CRMM-R).

The evaluation methodology consists of four primary components. First, the predictive capability of the proposed architecture is assessed using classification accuracy on the Medical Visual Question Answering task. Second, robustness experiments evaluate model performance under corrupted visual inputs to analyze resilience against noisy imaging conditions. Third, calibration experiments measure the reliability of predicted confidence scores using Expected Calibration Error (ECE). Finally, model compression experiments investigate the effect of dynamic quantization on predictive performance and deployment efficiency. Collectively, these experiments provide a comprehensive evaluation of both algorithmic performance and practical applicability.

A Experimental Environment

All experiments were implemented using the **PyTorch deep learning framework**, which provides efficient support for multimodal neural network development, automatic differentiation, GPU acceleration, and model optimization. Image preprocessing operations were performed using the **TorchVision** library, while numerical computations were carried out using **NumPy**. Statistical analysis and result visualization were conducted using **Pandas** and **Matplotlib**, respectively.

The complete implementation was developed in **Python**, allowing modular integration of the image encoder, question encoder, cross-modal attention mechanism, and classification network into a unified training pipeline. The implementation emphasizes reproducibility through consistent preprocessing procedures, deterministic data loading, and standardized evaluation protocols.

B Hardware Configuration

Training and evaluation were performed on a resource-constrained computing platform representative of practical deployment environments. The experiments were executed using a standard personal computing system rather than specialized high-performance servers, reflecting the deployment objective of CRMM-R.

The hardware configuration included:

Component	Specification
Processor	Apple Silicon CPU
GPU	Apple Integrated GPU(MPS backend)
Memory	8 GB Unified Memory
Operating System	macOS

Framework	PyTorch

C Software Configuration

The experimental software environment is summarized in Table 9.

Table 9. Software Environment

Software Component	Version/Framework
Programming Language	Python
Deep Learning Framework	PyTorch
Image Processing	TorchVision
Numerical Computing	Numpy
Data Analysis	Pandas
Visualization	Matplotlib
Operating System	macOS

D Dataset Preparation

The experiments utilize the publicly available **VQA-RAD** benchmark dataset consisting of paired medical images and clinical questions. Prior to training, all samples undergo the preprocessing pipeline described in Section IV-C.

Medical images are resized to a fixed spatial resolution of **128 × 128 pixels**, converted into tensor representations, and normalized prior to feature extraction. Clinical questions are tokenized, mapped to a predefined vocabulary, and padded to a fixed sequence length. Ground-truth answers are transformed into categorical class labels using label encoding, thereby formulating Medical Visual Question Answering as a supervised multi-class classification problem.

Mini-batch processing is performed using the PyTorch **DataLoader**, which shuffles training samples before each epoch to improve statistical diversity during optimization.

E Hyperparameter Configuration

To ensure consistent comparison across all experiments, identical hyperparameters are employed for both CRMM v1 and the proposed CRMM-R framework.

Table 10. Hyperparameter Configuration

Hyperparameter	Value
Batch Size	16
Image Resolution	128 x 128
Learning Rate	0.001
Optimizer	Adam
Loss Function	Cross-Entropy Loss
Number od Epochs	15
Image Feature Dimension	256
Text Feature Dimension	256
Activation Function	ReLU
Output Function	Softmax

The selected hyperparameters provide a balance between stable optimization, computational efficiency, and convergence speed while maintaining compatibility with resource-constrained hardware.

F Training Strategy

The proposed multimodal framework is trained using supervised learning. During each training iteration, a mini-batch containing synchronized medical images, encoded clinical questions, and corresponding answer labels is forwarded through the complete CRMM-R architecture.

The CNN image encoder extracts visual representations from the medical images, while the GRU encoder generates contextual embeddings from the associated clinical questions. These modality-specific representations are subsequently integrated using the proposed Cross-Modal Attention Module before being forwarded to the classification network.

Model parameters are optimized using the **Adam optimization algorithm**, which adaptively adjusts learning rates for individual parameters based on first- and second-order gradient estimates. Cross-Entropy Loss serves as the optimization objective, encouraging higher prediction probabilities for correct answer classes while penalizing incorrect predictions.

Training continues for multiple epochs until convergence, with model evaluation performed after each epoch to monitor predictive performance.

G Evaluation Metrics

Unlike conventional image classification studies that rely exclusively on predictive accuracy, the proposed work evaluates multiple complementary performance dimensions to obtain a comprehensive assessment of multimodal reasoning capability.

The primary evaluation metrics include:

Classification Accuracy

Measures the percentage of correctly predicted answers over the complete evaluation dataset.

Noise Robustness

Evaluates model resilience under corrupted visual inputs by comparing predictive accuracy between clean images and images corrupted with Gaussian noise and blur perturbations.

Expected Calibration Error (ECE)

Measures the agreement between prediction confidence and actual prediction correctness. Lower ECE values indicate better confidence calibration and more reliable probability estimates.

Compression Sensitivity

Assesses the effect of dynamic post-training quantization on predictive accuracy. This experiment evaluates whether storage reduction can be achieved without significant degradation in model performance.

Failure Analysis

Investigates incorrect predictions to identify systematic reasoning failures, including textual bias, weak visual grounding, dataset shortcut learning, and high-confidence misclassification.

Resource Efficiency

Evaluates deployment characteristics including:

- Number of trainable parameters
- Model storage size
- Quantized model size
- CPU inference latency
- Throughput
- Edge deployment suitability

These complementary metrics provide a multidimensional evaluation of CRMM-R beyond conventional classification accuracy.

VI. Experimental Results and Analysis

This section presents a comprehensive evaluation of the proposed **Cross-Modal Resource-Efficient Multimodal Reasoning (CRMM-R)** framework using the experimental protocol described in Section V. The evaluation investigates the effectiveness of the proposed architecture from multiple perspectives, including predictive accuracy, multimodal reasoning capability, robustness against noisy visual inputs, confidence calibration, compression sensitivity, computational efficiency, and qualitative failure analysis. Collectively, these experiments provide a holistic assessment of the proposed framework under realistic deployment conditions.

Unlike conventional Medical Visual Question Answering studies that primarily report classification accuracy, this work adopts a broader evaluation strategy by incorporating reliability and efficiency metrics that are particularly relevant for resource-constrained healthcare applications. Such comprehensive evaluation enables analysis not only of predictive performance but also of the trustworthiness and deployability of the proposed system.

A Baseline Performance Evaluation

The primary objective of the first experiment is to evaluate the effectiveness of the proposed Cross-Modal Attention Fusion Module by comparing it against the baseline multimodal fusion strategy. The baseline architecture, referred to as **CRMM v1**, combines image and question features through direct feature concatenation prior to classification. In contrast, the proposed **CRMM-R** replaces feature concatenation with the lightweight Cross-Modal Attention Module described in Section IV-F.

Both architectures were trained under identical experimental conditions using the same dataset, preprocessing pipeline, optimization strategy, and hyperparameter configuration. Consequently, any observed improvement can be directly attributed to the proposed multimodal fusion mechanism rather than differences in training methodology.

Table 11. Performance Comparison Between CRMM v1 and CRMM-R

Model	Fusion Strategy	Accuracy
CRMM v1	Feature Concatenation	28.20%
CRMM-R(Proposed)	Cross-Modal Attention	35.81%

The experimental results demonstrate a substantial improvement in predictive performance following the introduction of the proposed Cross-Modal Attention Module. The classification accuracy increases from **28.20%** to **35.81%**, corresponding to an absolute improvement of **7.61 percentage points**. This improvement confirms that explicit multimodal interaction enables more effective reasoning than independent feature concatenation.

Unlike the baseline architecture, which relies on the classifier to implicitly discover relationships between image and textual features, the proposed attention mechanism explicitly models these relationships before classification. Consequently, the network learns stronger semantic alignment between medical images and clinical questions, allowing the classifier to operate on richer multimodal representations.

These findings validate the central hypothesis of this work: **lightweight cross-modal attention can substantially improve multimodal reasoning while maintaining computational efficiency suitable for resource-constrained deployment**.

B Calibration Analysis

Reliable confidence estimation is essential for clinical decision-support systems because prediction confidence often influences the degree of trust placed in automated recommendations. High predictive accuracy alone does not guarantee reliable decision making if the predicted confidence values do not accurately reflect actual correctness.

Calibration quality is therefore evaluated using **Expected Calibration Error (ECE)**, which measures the discrepancy between prediction confidence and observed prediction accuracy. Lower ECE values indicate better alignment between confidence estimates and actual model performance.

Table 13. Calibration Results

Metric	Observed Value	Interpretation
Expected Calibration Error (ECE)	**0.0437**	Excellent calibration

The observed ECE value of **0.0437** indicates that the predicted confidence scores closely correspond to actual prediction accuracy. This result suggests that the proposed framework produces reliable probability estimates rather than systematically overestimating or underestimating prediction certainty.

Well-calibrated confidence estimates are particularly valuable in healthcare because they enable clinicians to distinguish between highly reliable predictions and cases requiring additional expert review. Consequently, the proposed CRMM-R framework offers improved practical reliability beyond conventional classification accuracy.

C Model Compression Analysis

Deployment in resource-constrained environments requires compact neural networks that minimize storage and computational requirements while preserving predictive performance. To evaluate deployment readiness, dynamic post-training quantization was applied to the trained CRMM-R model.

Table 14. Model Compression Results

Metric	Before Quantization	After Quantization
Model Size	**5.23 MB**	**0.90 MB**
Accuracy	**0.28**	**0.35**
Accuracy Drop	—	*0.07*

The experimental results demonstrate that dynamic quantization substantially reduces model storage requirements while introducing only minimal degradation in predictive performance. Such behavior indicates that the learned multimodal representations remain stable despite reduced numerical precision.

These findings confirm that CRMM-R is suitable for deployment on embedded healthcare systems where storage capacity and computational resources are limited.

D Resource Efficiency Analysis

Beyond predictive performance, computational efficiency constitutes an essential evaluation criterion for resource-constrained Medical VQA systems.

Table 15. Resource Utilization

Metric	Value
Trainable Parameters	~1.37 Million

Model Size	5.23 MB
Quantized Model Size	0.90 MB
CPU Inference Time	353.89 ms/batch
Throughput	45.21 images/sec

The proposed framework achieves an effective balance between predictive capability and computational efficiency. Compared with large transformer-based multimodal architectures, CRMM-R significantly reduces parameter complexity while maintaining practical reasoning capability, making it well suited for edge deployment.

E Failure Analysis

To investigate the limitations of the proposed framework, incorrectly classified samples were manually examined following model inference. Misclassified instances were analyzed according to prediction confidence, semantic consistency, and multimodal reasoning behavior.

Table 16. Observed Failure Categories

Failure Category	Observation	Possible Cause
Textual Bias	Prediction dominated by question semantics	Weak visual grounding
Weak Visual Evidence	Incorrect interpretation of subtle anatomical structures	Limited CNN representation

Dataset Shortcut Learning	Frequent answers predicted regardless of image content	Dataset imbalance
High-Confidence Errors	Incorrect predictions with strong confidence	Imperfect confidence estimation
Ambiguous Clinical Questions	Multiple plausible interpretations	

The qualitative analysis reveals that the majority of prediction errors originate from insufficient visual grounding rather than failures of textual understanding. Several high-confidence misclassifications indicate that although calibration remains strong overall, certain challenging clinical scenarios continue to expose limitations in multimodal reasoning. These observations motivate future research on stronger visual encoders, richer attention mechanisms, and improved multimodal alignment strategies.

F Overall Performance Summary

Table 17. Overall Experimental Summary

Experiment	Metric	Result
Baseline Accuracy	CRMM v1	**28.20%**
Proposed Accuracy	CRMM-R	**35.81%**
Accuracy Improvement	Absolute Gain	**+7.61%**
Calibration	Expected Calibration Error	**0.0437**

Noise Robustness	Gaussian / Blur	**Clean Accuracy: 0.2820284697508897** **Noisy Accuracy: 0.28247330960854095** **Drop: -0.000444839857651258**
Model Compression	Quantized Model Size	**0.90 MB**
Resource Efficiency	Parameters	**1.37 M**
CPU Throughput	Images/sec	**45.21**

VII. Discussion

The experimental results presented in Section VI demonstrate that the proposed **Cross-Modal Resource-Efficient Multimodal Reasoning (CRMM-R)** framework effectively addresses the primary research objective of improving multimodal reasoning for Medical Visual Question Answering while maintaining computational efficiency suitable for resource-constrained deployment. The observed improvements across predictive accuracy, confidence calibration, and model compactness collectively indicate that explicit cross-modal interaction provides measurable advantages over conventional feature concatenation without introducing substantial computational overhead.

One of the most significant findings of this study is the performance improvement achieved through the proposed **Cross-Modal Attention Fusion Module**. Replacing direct feature concatenation with question-guided attention increased classification accuracy from **28.20%** to **35.81%**, corresponding to an absolute improvement of **7.61 percentage points**. This result suggests that the quality of multimodal interaction is more influential than merely increasing model complexity. Rather than independently extracting visual and textual representations, the proposed attention mechanism establishes semantic correspondence between both modalities before classification, enabling the model to focus selectively on diagnostically relevant visual evidence. These observations support the growing consensus within multimodal learning literature that explicit feature interaction is essential for effective cross-modal reasoning.

The qualitative behavior of the attention mechanism further illustrates its contribution to multimodal reasoning. Medical Visual Question Answering differs fundamentally from conventional image classification because the correct interpretation of a medical image depends on the accompanying clinical question. A single radiological image may contain numerous anatomical structures, yet only a subset is relevant to a particular query. By allowing the textual representation to guide visual feature selection, the proposed architecture performs adaptive reasoning instead of relying on fixed feature representations. Consequently, identical medical images can produce different internal attention patterns depending on the clinical question being posed, thereby improving semantic consistency during answer prediction.

Another important observation concerns the relationship between **predictive accuracy and confidence calibration**. Although the overall classification accuracy remains constrained by the limited size and diversity of the VQA-RAD dataset, the framework demonstrates an **Expected Calibration Error (ECE) of 0.0437**, indicating close agreement between predicted confidence and actual prediction correctness. This finding is particularly significant because confidence calibration and predictive accuracy represent distinct aspects of model behavior. A model may achieve moderate accuracy while still producing highly reliable probability estimates if its confidence accurately reflects prediction uncertainty. From a clinical perspective, well-calibrated confidence estimates are essential because they enable healthcare professionals to distinguish between predictions that can be trusted with greater certainty and those requiring additional clinical verification. Therefore, the proposed framework contributes not only to predictive performance but also to the reliability and interpretability of automated decision support.

The resource efficiency analysis further demonstrates that strong multimodal reasoning does not necessarily require computationally intensive transformer-based architectures. The complete CRMM-R framework contains approximately **1.37 million trainable parameters**, occupies only **5.23 MB** of storage, and can be compressed to **0.90 MB** through dynamic quantization with minimal degradation in predictive performance. These characteristics highlight the feasibility of deploying multimodal reasoning systems on embedded hardware, portable medical devices, and edge computing platforms where memory capacity and computational resources are inherently limited. The successful preservation of predictive performance following quantization also indicates that the learned feature representations possess substantial numerical robustness, making the framework well suited for low-precision inference.

The robustness experiments provide additional insight into the strengths and limitations of the proposed architecture. Performance degradation under noisy image conditions remained comparatively moderate, suggesting that the multimodal attention mechanism reduces excessive dependence on individual visual features by incorporating complementary contextual information from the accompanying clinical question. Nevertheless, qualitative analysis of incorrectly classified samples indicates that certain challenging diagnostic cases continue to expose limitations in visual representation learning. Several failure cases were associated with subtle anatomical abnormalities, low-contrast structures, or visually ambiguous imaging regions, indicating that the lightweight CNN encoder may not always capture sufficiently discriminative representations for highly complex clinical scenarios.

Failure analysis also reveals evidence of **textual bias** and **dataset shortcut learning**, phenomena frequently reported in Medical VQA literature. In several cases, the model produced predictions that appeared to rely more heavily on statistical correlations within the question distribution than on explicit visual evidence. This behavior is likely influenced by the relatively small size of the VQA-RAD dataset and the imbalance in answer frequencies, which may encourage the learning of dataset-specific shortcuts rather than generalized multimodal reasoning strategies. Although the proposed cross-modal attention mechanism reduces this tendency compared with simple feature concatenation, it does not eliminate it entirely. Future research should therefore investigate stronger visual grounding techniques and larger, more diverse medical VQA datasets to further improve multimodal reasoning robustness.

From a broader perspective, the findings of this study demonstrate that **efficient architectural design** can provide a practical alternative to increasingly large multimodal foundation models. Contemporary vision-language systems frequently prioritize predictive performance by substantially increasing model size, parameter count, and computational complexity. While these approaches often achieve state-of-the-art accuracy, their deployment remains challenging in many real-world healthcare environments. In contrast, the proposed CRMM-R framework illustrates that carefully designed lightweight architectures incorporating efficient cross-modal attention can achieve meaningful improvements in multimodal reasoning while preserving deployment feasibility. This balance between predictive capability and computational efficiency represents an important consideration for the future development of clinically deployable artificial intelligence systems.

The overall findings therefore support the central hypothesis established at the beginning of this study: **explicit cross-modal interaction provides greater benefit to multimodal reasoning than simply increasing architectural complexity**. By integrating efficient attention-based feature alignment with a compact network architecture, CRMM-R successfully improves predictive performance, maintains reliable confidence estimation, supports model compression, and satisfies the practical computational requirements associated with resource-constrained healthcare applications. These characteristics collectively position the proposed framework as a promising direction for future research in efficient Medical Visual Question Answering.

VIII. Future Enhancements

Although the proposed **Cross-Modal Resource-Efficient Multimodal Reasoning (CRMM-R)** framework demonstrates promising performance for Medical Visual Question Answering under resource-constrained environments, several opportunities exist to further enhance its predictive capability, generalization, and clinical applicability. The experimental findings presented in this study identify multiple research directions that can strengthen multimodal reasoning while preserving the lightweight characteristics of the proposed architecture. These future enhancements encompass improvements in dataset diversity, representation learning, multimodal fusion, model interpretability, deployment strategies, and clinical integration.

One of the primary directions for future research involves **expanding the training dataset**. The VQA-RAD dataset, while widely adopted for benchmarking Medical VQA systems, contains a relatively limited number of image-question pairs and represents only a subset of clinical imaging scenarios. Such limitations restrict the diversity of visual patterns and clinical concepts encountered during training, thereby constraining the model's generalization capability. Future work may incorporate larger publicly available datasets such as **PathVQA**, **SLAKE**, or other multimodal medical benchmarks to expose the model to broader anatomical structures, disease categories, imaging modalities, and clinical question types. Training on more diverse datasets is expected to improve robustness and reduce overfitting to dataset-specific characteristics.

Another promising enhancement involves the development of **stronger visual feature extraction mechanisms**. The lightweight convolutional neural network employed in CRMM-R successfully balances computational efficiency and predictive performance; however, qualitative failure analysis indicates that subtle anatomical abnormalities and fine-grained pathological structures remain challenging to recognize. Future implementations may investigate hybrid architectures that combine compact convolutional networks with lightweight Vision Transformer (ViT) modules or hierarchical attention mechanisms to improve spatial representation learning while maintaining acceptable computational complexity.

The proposed Cross-Modal Attention Module may also be extended through **multi-head cross-attention mechanisms**. The current architecture employs a lightweight attention strategy to preserve efficiency, yet multiple parallel attention heads could enable simultaneous learning of complementary visual-textual relationships at different semantic levels. Such extensions may improve reasoning for complex clinical questions involving multiple anatomical regions or pathological findings while providing richer multimodal feature representations.

Future research may additionally explore the integration of **pre-trained biomedical language models** for enhanced question understanding. Although the GRU encoder provides efficient sequential representation learning, domain-specific language models trained on large-scale biomedical literature, such as **BioBERT**, **ClinicalBERT**, or **PubMedBERT**, possess substantially richer semantic representations of medical terminology and clinical concepts. Incorporating these language models into a lightweight multimodal architecture may improve comprehension of complex clinical questions without fundamentally altering the proposed multimodal reasoning framework.

Another important direction involves incorporating **medical knowledge graphs and external clinical knowledge bases**into the reasoning process. Current Medical VQA models primarily rely on statistical relationships learned directly from training data, which may limit their ability to reason about rare diseases or uncommon clinical conditions. Integrating structured medical knowledge could enable the framework to perform knowledge-guided reasoning by combining learned visual representations with established clinical relationships, thereby improving both prediction accuracy and interpretability.

The experimental analysis also highlights the importance of **visual explainability** for clinical deployment. Although the proposed attention mechanism implicitly identifies relevant image regions, future work should incorporate explicit explainability techniques such as **Grad-CAM**, **attention heatmaps**, or **saliency visualization**. These approaches would allow clinicians to inspect the anatomical regions influencing model predictions, thereby increasing transparency, supporting clinical validation, and improving user confidence in automated decision-support systems.

The deployment-oriented nature of CRMM-R also motivates future research on **federated learning** for distributed healthcare environments. Medical institutions often face strict privacy regulations that restrict centralized sharing of patient data. Federated learning enables collaborative model training across multiple hospitals while keeping sensitive patient information locally stored. Integrating federated optimization into CRMM-R would facilitate continual learning from geographically distributed medical datasets while preserving patient confidentiality and regulatory compliance.

Further improvements may also focus on **adaptive model compression techniques** beyond dynamic quantization. Approaches such as structured pruning, knowledge distillation, mixed-precision inference, neural architecture search, and hardware-aware optimization may further reduce computational requirements while preserving predictive performance. Such techniques would enable deployment on increasingly constrained hardware platforms, including wearable medical devices and Internet of Medical Things (IoMT) systems.

Future investigations should additionally evaluate the proposed framework under **multi-modal clinical settings**, where Medical Visual Question Answering extends beyond images and textual questions to incorporate electronic health records, laboratory reports, radiology findings, genomic information, and physiological signals. Such multimodal integration would enable richer clinical reasoning and provide more comprehensive decision-support capabilities for real-world healthcare applications.

Finally, large-scale **prospective clinical validation** remains an essential step toward real-world deployment. Although the proposed framework demonstrates encouraging performance on benchmark datasets, evaluation using diverse patient populations, multiple healthcare institutions, and real clinical workflows is necessary to establish reliability, robustness, and clinical utility. Such studies would facilitate translation of lightweight multimodal reasoning systems from controlled research environments into practical healthcare applications.

Collectively, these future enhancements illustrate that CRMM-R represents a flexible and extensible foundation for efficient Medical Visual Question Answering rather than a fixed architecture. The modular design of the framework allows new visual encoders, language models, attention mechanisms, compression strategies, and knowledge integration techniques to be incorporated without fundamentally redesigning the complete system. Consequently, the proposed architecture provides a scalable platform for continued research toward accurate, interpretable, and resource-efficient multimodal clinical intelligence.

IX. Conclusion

Medical Visual Question Answering has emerged as a promising research area at the intersection of computer vision, natural language processing, and intelligent clinical decision support. Despite significant progress in multimodal representation learning, many existing Medical VQA systems remain computationally expensive, making their deployment challenging in resource-constrained healthcare environments such as portable diagnostic devices, rural healthcare facilities, and edge computing platforms. Addressing this challenge requires multimodal architectures that balance predictive performance with computational efficiency while maintaining reliable reasoning capability.

This research presented **CRMM-R (Cross-Modal Resource-Efficient Multimodal Reasoning)**, a lightweight multimodal framework specifically designed for Medical Visual Question Answering under resource-constrained settings. The proposed architecture integrates a compact convolutional neural network for visual feature extraction, a Gated Recurrent Unit for contextual language representation, and a lightweight Cross-Modal Attention Fusion Module that enables explicit semantic interaction between medical images and clinical questions. Unlike conventional feature concatenation approaches, the proposed attention mechanism dynamically aligns visual and textual representations before classification, allowing the model to focus on clinically relevant visual evidence while preserving computational efficiency.

Comprehensive experimental evaluation demonstrated the effectiveness of the proposed framework across multiple performance dimensions. The Cross-Modal Attention Module improved classification accuracy from **28.20%** in the baseline architecture to **35.81%**, corresponding to an absolute improvement of **7.61 percentage points**. In addition to improved predictive performance, the framework achieved an **Expected Calibration Error (ECE) of 0.0437**, indicating reliable confidence estimation suitable for decision-support applications. Resource-oriented evaluation further confirmed the efficiency of the proposed design, with approximately **1.37 million trainable parameters**, a compact **5.23 MB** model size, successful post-training quantization to **0.90 MB**, and efficient CPU inference capable of supporting practical deployment scenarios.

Qualitative failure analysis provided additional insight into the strengths and limitations of the proposed framework. While the Cross-Modal Attention Module substantially improved semantic alignment between visual and textual modalities, several challenging cases revealed limitations associated with subtle anatomical structures, dataset imbalance, and incomplete visual grounding. These observations highlight the importance of stronger visual representations, larger multimodal medical datasets, and richer cross-modal reasoning mechanisms for future research.

Beyond quantitative performance improvements, this study demonstrates that **effective multimodal reasoning does not necessarily require computationally intensive architectures**. Careful architectural design, efficient feature interaction, and deployment-aware optimization can collectively produce Medical VQA systems that remain accurate, reliable, and computationally practical. This finding is particularly significant for healthcare applications where computational resources, storage capacity, and inference latency are critical deployment considerations.

Overall, the proposed CRMM-R framework establishes a practical foundation for resource-efficient Medical Visual Question Answering by combining lightweight neural architectures with explicit cross-modal semantic reasoning. The framework contributes to the growing body of research on deployable multimodal artificial intelligence and demonstrates that efficient attention-based feature interaction can substantially improve reasoning performance without sacrificing deployment feasibility. Future extensions incorporating larger medical datasets, advanced visual encoders, biomedical language models, explainable artificial intelligence techniques, and knowledge-guided multimodal reasoning are expected to further enhance both predictive capability and clinical applicability. Consequently, CRMM-R represents an important step toward the development of scalable, trustworthy, and resource-aware intelligent systems capable of supporting next-generation clinical decision-making across diverse healthcare environments.

Appendix

Appendix A. Network Hyperparameters

The hyperparameters used during training were selected to balance predictive performance and computational efficiency while maintaining compatibility with resource-constrained hardware.

Table A1. Hyperparameter Configuration

Hyperparameter	Value
Image Resolution	128 × 128
Batch Size	16

Learning Rate	0.001
Optimizer	Adam
Loss Function	Cross-Entropy Loss
Number of Epochs	15 *(update if different)*
CNN Feature Dimension	256
GRU Hidden Dimension	256
Activation Function	ReLU
Output Activation	Softmax

Appendix B. Training Curves

Example Predictions

Representative prediction examples are included to illustrate the reasoning capability of the proposed framework.

Table E1. Sample Predictions

Medical Image	Clinical Question	Ground Truth	Predicted Answer	Prediction Status

Sample 1	Example question	Yes	Yes	Correct
Sample 2	Example question	No	Yes	Incorrect
Sample 3	Example question	Left Lung	Left Lung	Correct

Ablation Study

This appendix summarizes the contribution of the proposed Cross-Modal Attention Module.

Table G1. Ablation Results

Model Variant	Fusion Strategy	Accuracy (%)	Improvement
CRMM v1	Feature Concatenation	28.20	—
CRMM-R	Cross-Modal Attention	35.81	+7.61

This comparison demonstrates that replacing direct feature concatenation with cross-modal attention yields a measurable improvement in multimodal reasoning performance.

Appendix H. Resource Utilization

To evaluate deployment feasibility, computational efficiency metrics are summarized below.

Table H1. Resource Analysis

Metric	Value
Trainable Parameters	~1.37 Million
Model Size	5.23 MB
Quantized Model Size	0.90 MB
CPU Inference Time	353.89 ms/batch
Throughput	45.21 images/sec

These results demonstrate that CRMM-R is designed not only for improved multimodal reasoning but also for practical deployment in environments with limited computational resources.

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Code Availability

The implementation of CRMM-R, including the training pipeline, evaluation scripts, and experimental configurations, is publicly available at:

GitHub:https://github.com/Karthika1403/On-Device-Multimodal-Health-Reasoning-Under-Resource-Constraints

Google-Colab-Notebook:
https://colab.research.google.com/drive/1HuaMN0u6Avn11wxUfYZY9bsw1dteTvZX